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Sign Language Recognition using Python

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Abstract—This paper presents a real-time computer vision and machine learning-based hand gesture recognition system using MediaPipe Hands for landmark detection and a Random Forest Classifier for gesture classification. The system achieves 93.6% test accuracy in recognizing static ASL alphabets (A-Z) through a pipeline comprising data collection, feature extraction, model training, and real-time deployment. While demonstrating robustness for static gestures, limitations include difficulty with dynamic signs and multi-hand inputs. The system's affordability and efficiency make it suitable for applications in human-computer interaction and assistive technology.

Index Terms—Sign Language Recognition, MediaPipe, Random Forest, Real-time Systems, Computer Vision

I. INTRODUCTION

Gesture recognition has emerged as a vital component of human-computer interaction, particularly in sign language interpretation. Approximately 70 million people globally use sign language as their primary means of communication [1], yet technological solutions remain limited by cost and accuracy constraints.

A. Background

Early sign language recognition (SLR) systems relied on sensor gloves [2], while modern approaches leverage computer vision and machine learning. MediaPipe Hands [3] has revolutionized real-time landmark detection, enabling markerless gesture recognition.

B. Problem Statement

Existing SLR solutions face three key challenges:

- High hardware requirements.
- Limited accuracy (typically <90% for complex gestures).
- Inability to process dynamic signs.

Our work addresses these through an efficient vision-based system using consumer-grade hardware.

C. Contributions

This paper contributes:

- A complete ML pipeline for static ASL recognition.
- Empirical evaluation showing 93.6% test accuracy.
- Open-source implementation for community adoption.

II. RELATED WORK

Recent advances in SLR can be categorized as follows:

A. Sensor-Based Approaches

Dipietro et al. [2] demonstrated glove-based systems achieving 99% accuracy but requiring specialized hardware.

B. Vision-Based Methods

MediaPipe Hands [3] provides 21 landmarks per hand with 95% detection accuracy at 30 FPS. Chen et al. [4] combined landmarks with geometric features for improved recognition.

C. Classification Techniques

Random Forests achieve 96.66% accuracy for ISL [5], while CNNs excel in complex scenarios [6] at higher computational cost.

III. METHODOLOGY

Our system architecture (Fig. 1) comprises four stages:

A. Data Collection

Custom dataset of 2,600 images (100 per ASL letter), 80/20 train-test split with stratification, and variations in lighting and orientation.

B. Feature Extraction

MediaPipe Hands extracts 21 normalized landmarks (Fig. 2). The feature vector is defined as:

$$\text{Feature Vector} = [x_1, y_1, z_1, \dots, x_{21}, y_{21}, z_{21}]$$

C. Model Training

Random Forest classifier with:

- 100 decision trees.
- Gini impurity criterion.
- 5-fold cross-validation.

D. Real-Time Deployment

OpenCV pipeline processes webcam feed at 30 FPS with:

- Landmark detection (8ms/frame).
- Feature normalization (2ms).
- Classification (3ms).

TABLE I
MODEL ACCURACY METRICS

Metric	Value
Training Accuracy	98.2%
Test Accuracy	93.6%
Precision	92.8%
Recall	93.1%

IV. RESULTS

A. Accuracy Metrics

B. Confusion Analysis

Common misclassifications (Fig. 3):

- $F \leftrightarrow G$ (23% error rate).
- $K \leftrightarrow V$ (18% error rate).
- Dynamic signs (J, Z) unrecognized.

C. Limitations

- Single-hand restriction.
- No unknown gesture detection.
- 12% latency increase in low light.

V. CONCLUSION

We present a real-time ASL recognition system achieving 93.6% accuracy using consumer hardware. Future work will incorporate LSTM networks for dynamic gestures and expand the gesture vocabulary.

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REFERENCES

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